

The influence of organizational culture on the adoption of extreme programming

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Received 1 February 2007; received in revised form 11 January 2008; accepted 19 January 2008

Available online 1 February 2008

Abstract

The adoption of extreme programming (XP) method requires a very peculiar cultural context in software development companies. However, stakeholders do not always consider this matter and tend to stand to technical requirements of the method. Hence this paper aims at identifying aspects of organizational culture that may influence favorably or unfavorably the use of XP. In order to identify those aspects, this study analyzes dimensions of organizational culture under the perspective of practices and values of XP. This paper is based on the review of the literature of the area and empirical observations carried out with six software companies. This study does not intend to develop a tool for measurement of XP's compatibility with the organizational culture of each company. It intends to provide parameters (favorable and unfavorable aspects) for previous consideration of the convenience of XP implementation.

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Keywords: Software development companies; Organizational culture; Extreme programming

1. Introduction

The agile software development method extreme programming – XP (Beck, 2000) promises high performance to teams adopting it throughout. However, satisfying technical aspects does not suffice to implement extreme programming. It is fundamental that some cultural aspects should be present at the working environment (Larsen and Kerievsky, 2003).

As regards to implementing new methods, Mcbreen (2002) warns that adaptation to a new software development method is not always an easy task, since organization members need to change attitude and values. To McBreen, changes on organizational culture are intriguing, but rarely considered. Henceforth, it is necessary to choose a software development method that suits the organizational culture.

Within some cultural aspects necessary for the implementation of XP, the following can be pinpointed: company policies related to employees' motivation, cohesion in the development team, and democratic leadership. The absence of these aspects at the organizational environment may demand deep changes in the company. In the same way, changes may be necessary in case there are certain aspects hindering XP implementation, as for example, predominance of individual work, lack of communication and integration among software developers, and company resistance in relation to changes (e.g. adoption of new technologies).

It is important to note that if the company interested in the adoption of XP already uses an agile method (e.g. SCRUM, Schwaber, 1985), the impact may be smaller. This happens because XP requires that some cultural aspects (also required by other agile methods) already are part of the environment. Otherwise, the implementation of the agile method would not be successful. Nevertheless, XP is often the first agile method that companies intend to implement.

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Within this context, a concern arises. This concern has to do with the wearing out originated from XP implementation in companies which organizational culture is highly incompatible with practices and values of the method. Thus this study aims at highlighting cultural aspects of software companies, which may influence favorably or unfavorably XP adoption. This may provide parameters to enable prior consideration over the method's compatibility with the cultural organization of the company.

Based on the favorable and unfavorable aspects presented in this paper, people interested in adopting XP can investigate whether the company is more or less compatible with the organizational culture required by XP. This paper does not present a tool for measuring compatibility, but it provides parameters for a previous assessment of this compatibility.

2. Extreme programming

Extreme programming is one of the software development methods that has most stood out among the so-called agile methods (Agile, 2001). Its premises are based on practices and values that complement each other, and they should be carefully followed for the effectivity of the method.

XP guiding values are: communication, simplicity, feedback, and courage (Beck, 2000). These values should be present in the practices prescribed by the method, which are: On-site Customer, Planning Game, Metaphor, Simple Design, Small Release, Pair Programming, Testing, Refactoring, Continuous Integration, 40-Hour Week, Coding Standard, and Collective Ownership (Beck, 2000).

According to Abrahamsson et al. (2003), the Software Engineering discipline already recognizes and uses most of the XP method, such as testing and refactoring. Even so, XP method has a particularity, i.e. need for synergy among its practices, which complete each other.

Furthermore, it is important to notice that XP paradigm differs from traditional software development methods in the sense that it is deeply focused on human relationships. In other words, XP conception and evolution depend on acceptance, compromise and interactions among people, who are, in this case, stakeholders. According to Xexéo (2006), stakeholders are all those interested in the software system, i.e. those who affect or are affected by XP results.

Astels et al. (2002) and Beck (2000), similarly to greater part of the literature, do not establish rigidly stakeholders' roles/functions. On the other hand, authors who deal with the matter generally classify roles of people involved with an XP project in two groups: (i) those that form the software development team, i.e. developers, and (ii) those representing the customer.

Users and those with enough expertise to interact with developers to define and approve the system may represent the customer. To Astels et al. (2002), investors and project resource suppliers are also part of the group that represents the client. Jeffries (1999) denominates them "gold owners".

People who are part of the organizational structure, and somehow have influenced to adopt XP, such as managers and company managers, should also be considered stakeholders.

In terms of an XP project, it is important to note that a team member may be simultaneously involved in planning, analyzing, programming and testing. Henceforth this paper employs the term *developer* to refer to analysts, architects, programmers, testers, the ones responsible for quality, as well as whoever is involved in software development and conception.

3. Organizational culture

Chiavenato (1999) defines organizational culture as a system of meanings shared among all organization members, so that the organization may be distinguished from the rest. This has to do with the company's institutionalized way of thinking and acting. The essence of a company is expressed by the way it does business, the way it deals with its customers and employees, and the level of autonomy it grants to its members.

Schein (1985) defines organizational culture as a standard set of basic suppositions invented, discovered or developed by the group when learning to face problems of external adaptation and internal integration. These basic suppositions must work well enough so that they may be considered valid, and then be taught to new members as the correct way of perceiving, thinking and sensing problems.

Thus, changing the culture of an organization is not a trivial task, since it requires changes on the members' ways of thinking, communicating, relating to each other, as well as consolidated habits and working manners.

Furthermore, we want to highlight that in order to successfully program changes in companies (e.g. introducing a new working method), multiple faces (or dimensions) of organizational culture should be considered. This should be done because culture is a condition for the success of intended innovation.

Studying the theme, O'Reilly et al. (1991) identified seven characteristics (*dimensions*, in this paper) that may define the essence of organizational culture when taken as a whole. These dimensions are Innovation and Risk, Detail Orientation, Outcome Orientation, People Orientation, Team Orientation, Aggressiveness, and Stability.

According to Robbins (1999), these seven dimensions are somehow present at all companies, and their analyses bring forth comprehension to organizational culture. This occurs because these characteristics provide an image of how things are done at a company, and of the way its members are expected to behave. Table 1 shows the meaning of each dimension.

Each dimension has its own cultural aspects. These cultural aspects vary according to the level of characteristics each dimension establishes. For example, a cultural aspect of "Innovation and Risk" may be "having group members as innovators". Regarding "Detail Orientation", a

Table 1
Dimensions of organizational culture (adapted from Robbins, 1999)

Dimension	Description
Innovation and Risk	Corresponds to the level to which employees are motivated to be creative, to innovate, and to shoulder risks when taking decisions
Detail Orientation	Corresponds to the level expected to which employees demonstrate precision, analysis, and attention to details
Outcome Orientation	Corresponds to the level on which administration focuses more on results than on techniques and processes used to obtain them
People Orientation	Corresponds to the level on which decisions of the administration take into account their effects regarding people in the company
Team Orientation	Corresponds to the level on which work activities are done, having the team for base, rather than each member individually
Aggressiveness	Corresponds to the level on which people are more aggressive and competitive than cooperative
Stability	Corresponds to the level on which activities focus on maintaining <i>status quo</i>

constant product enhancement philosophy paying special attention to details could be a cultural aspect. In terms of “Team Orientation”, the cultural aspect may be “predominance of teamwork”, or maybe “individual work”. The same occurs to the other dimensions.

4. Analyses of software companies

Six software companies have been studied. Each of them received a fictional name, and only their main activity is reported (see Table 2).

Initially it is important to mention that research has been carried out with these companies according to two main purposes: (i) verify whether or not the related aspects are present in each organizational culture dimension of the company; and (ii) examine how these aspects may influence on the adoption of XP.

Cultural aspects of every dimension have been detected at all companies. Unfortunately, due to formal restrictions, we chose to comment only on the most prominent cultural aspects of each company studied. In this way, in the following session each company was mentioned in one of the dimensions of the organizational culture.

Table 2
List of companies analyzed

Fictional name	Company's main activity
Corporativa	Corporate web portal development
Desktop	Database development and commercial applications
Webgames	Desktop-game and web multiplayer development
Web	Webpage development, intranet, consultancy, e-business and e-commerce solutions
Desktop 2	Database development and commercial applications
Departamento	Software development department at a College

5. Dimensions of organizational culture under the perspective of extreme programming

Identifying favorable and unfavorable aspects for the adoption of XP method requires analysis of the dimensions of organizational culture under the perspective of XP practices and values. The way this is carried out is explained below.

This analytical activity enabled creating a checklist for each dimension of organizational culture. This list, which aims at revealing cultural aspects of organizations, is a series of questions made for companies.

The checklist applied to companies was composed by questionnaires, interviews, and on-site observations.

Thus, there is a checklist right after some considerations relative to each dimension of organizational culture. Furthermore, there are sceneries which purpose is to illustrate the empirical context in which the most prominent cultural aspects of each company were identified.

Last but not least, there is a list of favorable and unfavorable aspects to XP adoption after considering the aforementioned activities.

5.1. Dimension Innovation and Risks

The dimension *Innovation and Risks* corresponds to the level to which employees are motivated to innovate and shoulder risks (Robbins, 1999).

Migrating to XP method requires observing developers' disposition to innovate and shoulder risks, since considerable changes in the work routines established at the company may be necessary. Differently from more prescriptive software development methods, XP does not prescribe a detailed plan with all system requirements right at the beginning of the project. It assumes that requirements change during development.

As XP works with vague requirements, developers must be creative, innovative, skilled, and always prepared to unexpected events, since they do not have a long-term plan.

Furthermore, in XP method, customer participates actively of the software development process, having direct contact with his/her developers. Customer may alter requirements during implementation, introduce new stories, and even shift the direction of the project, considering that developers should be apt to deal with these sudden changes.

The iterative and incremental process used with XP emphasizes human characteristics of learning and making suppositions. In this way, developers are subject to mistakes when estimating, analyzing requirements, designing and testing, and error risks are unavoidable. To Cockburn (2001), many software managers resist to iterative and incremental software processes precisely because of the risk factor.

When asking members of a software company whether it is open to innovation the answer is probably positive. However, innovation may be just a value desired by the manager or by the team. It is possible to verify the

effectuation of these values observing mechanisms created to generate innovation, and also if the company members are potentially innovative.

Some attitudes that help to reveal whether employees have an innovative profile or not are identified by the habit of questioning whether there is a better way to do their activities, such as adapting ideas and putting them in their context of work, accepting routine changes, and expressing the desire of experimenting new technologies (Mayo, 2001).

On the other hand, in order to proportionate innovation and creativity, it is fundamental that the company be open to developers' suggestions, considering and implementing them. Developers need to sense that their opinions are important at work to feel motivated to develop software creatively.

The way developers deal with their own mistakes also strongly influence how much they are inclined to innovate. The company that usually "points at the guilty one", overtly reprehends, or even punishes those who fail, ends up inhibiting innovation, and developers' predisposition to shoulder risks. The way one deals with mistakes serves as a supporting factor of the company's existing culture. If culture has to do only with obeying norms and not shouldering risks, the fact that the system identifies, criticizes overtly, or even punishes "the guilty ones", assures permanence of this culture.

The rewarding system (money-wise or not) may also influence on the company members' disposition to work creatively and shoulder risks. Reward works as a positive reinforcement and signaler of the behavior the company expects from employees (Chiavenato, 2002). In this way, if the system is managed according to values such as the developers' creativity and performance, there will be a favorable environment to XP. Fig. 1 shows the checklist related to the dimension Innovation and Risk.

5.1.1. Scenario 1 – corporativa company

Some propositions mentioned previously have been confirmed at "Corporativa". At this company, the punishment system seemed to be one of the greatest impediments to the adoption of XP method.

The manager stated that he is an XP sympathizer, and that he was gradually adopting XP practices. According to him, an organized work environment was already being shared, and some XP practices had already been carried out, such as Refactoring, Pair Programming, and Continuous Integration.

At the company, we verified that there is in fact a shared environment, and code integration. However, code review is not frequent, and pair programming was limited to occasional support among colleagues, whenever one of them asked for.

The application of questionnaires revealed an important fact about identifying aspects of organizational culture. There was a large gap between developers' answers in the questionnaires, and what was in fact observed and affirmed in interviews. The question follows: How do developers' errors are handled when they may put the project in risk or retard the schedule? Answers obtained from the questionnaire are displayed in Fig. 2.

The questionnaire was applied to 10 developers at the company. Option (a) was predominant. It considers errors as part of the learning process.

However, when submitted to interview, developers reported a different reality. They demonstrated great discomfort in relation to managers' reaction towards developers' errors and mistakes. According to them, managers roughly criticize those who make errors before their colleagues. This established a constant preoccupation of not making errors, and restricts oneself to doing only what the manager determines.

When comparing questionnaire and interviews, discrepancies showed that the questionnaire does not portray the current panorama. This probably occurs because developers have chosen a less compromising answer, fearing a negative reaction on the part of their superiors.

This situation demonstrates that using only a questionnaire for analysis is not a safe option to identify organizational culture. Questionnaire should be used as an artifact for initial data collection, which should be compared with results obtained from other analytical instruments.

The aspects observed in that company "Corporativa" are according to what Beck (2000) suggests: an XP favorable work environment must provide developers with the freedom to give bad news to clients and managers, having no fear of punishment. When working at a system in which mistakes are punished, developers tend to work strictly based on the project plan, and feel reluctant to report the managers about doubts or identification of faults.

Error punishment reflects directly onto developers' performance. People act more carefully when they are not allowed to fail; they never surpass their limits, and tend to establish goals lower than those they can achieve (Tracy, 1994).

- | |
|---|
| <ol style="list-style-type: none"> 1. Do developers look for making suggestions to improve the software development process? 2. Do developers know how/have potential to innovate, shoulder responsibilities and risks? 3. Are developers' suggestions usually considered and implemented? 4. How are mistakes dealt with at the company? 5. Does the rewarding system encourage innovation? |
|---|

Fig. 1. Checklist items – dimension Innovation and Risk.

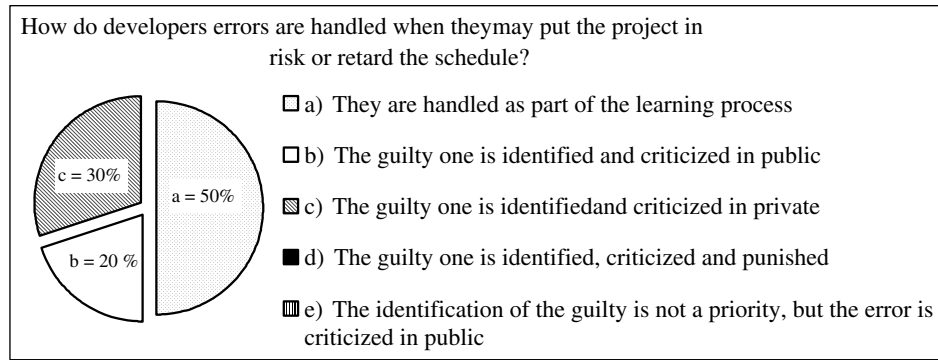


Fig. 2. Question number 7 of the questionnaire applied to developers.

At the company “Corporativa”, all developers share the same environment, being close enough to exchange information, and pair program (as required by XP). However, benefits obtained from developers’ sharing the same work environment was practically nullified due to the fact that the manager (punishing and controlling profile) shared the same work environment. In this way, the environment was constantly under surveillance, and the relationship among developers lacked informality and hindered communication.

This shows that, even if physical environment is favorable to the adoption of XP method, there are other aspects that may make it difficult. These aspects are identifiable through observations of the work environment.

5.1.2. Favorable and unfavorable cultural aspects for Innovation and Risk

Due to the remarks and reflections above, it is possible to identify the cultural aspects shown in Table 3 (favorable and unfavorable to the adoption of the XP) related to the dimension Innovation and Risk.

5.2. Dimension Detail Orientation

The dimension “Detail Orientation” corresponds to the level expected to which employees demonstrate precision, analysis, and attention to details (Robbins, 1999).

Detail-oriented is a basic premise to enable the adoption of XP. According to Burke and Coyner (2003), it is wrong

to think of XP as a chaotic software development methodology that pleases hackers. Actually, this method requires careful programmers who are detail-oriented and that are proud of their work.

Additionally, Asproni et al. (2004) also highlights that to be detail-oriented it is necessary to adopt agile methods, such as XP, and they require high standards of excellence. These patterns can be seen in the customer’s constant satisfaction with the delivery deadline and with the software, which must be functional, fully tested, simple, easy to modify, and with few or none errors.

The programmer should be careful by nature (Beck, 2000). There should be a philosophy of constant improvement; in a way that one may comprehend the whole without compromising the parts. XP programmers should be analytical and thorough, since the code must be exhaustively tested and reviewed.

The developer should be detail-oriented especially in order to: (i) Assure that the code is simple and comprehensive enough, so that the collective version may be safely refactored and altered, (ii) test intensively, (iii) understand the customer and help him create reports, as well applying acceptance tests, (iv) identify errors when pair programming, and (v) carry on continuous integration.

This dimension also implies iterative and incremental software development process with small releases, which is carried out with XP. This shows how much XP is focused on the software quality delivered to the customer, and it consequently requires more detail orientation in comparison to the traditional process.

In order to identify detail orientation in a software business company the importance given to the software should be observed. This can be seen by different methods: observing the team by using a programming pattern, observing whether there is refactoring, and also observing whether there is a formal plan for tests or any policy for software quality. Fig. 3 shows the checklist related to the dimension Detail Orientation.

5.2.1. Scenario 2 – “Desktop” company

One of the companies observed in this study lacked detail orientation, as presented below.

Table 3
Favorable and unfavorable aspects for Innovation and Risk

Dimension	Favorable aspects in relation to XP	Unfavorable aspects in relation to XP
Innovation and Risk	– Innovative and creative developers – Managers open to suggestions – Rewarding system which stimulates creativity and performance – Errors dealt with as part of the learning process	– Developers with a conservative profile – Managers closed to suggestions – Rewarding system stimulating only following orders – Error punishment system

1. Are quality, testing and reviews values shared by the company's members?
2. Which tests and reviews are carried out throughout the software development process?
3. Are developers careful and thorough in relation to each software module?

Fig. 3. Checklist items – dimension Detail Orientation.

At “Desktop”, interviews with managers and developers suggest that the greatest obstacles to put detail orientation in practice throughout software development process are: pressure referred to fulfillment of product deadline, shortage of financial resource, and lack of specialized workers. According to interviews, these aspects generally restrict initiatives for testing, reviewing, and improvement programs.

This company has two founding partners who are also administrators and project managers. At the company, there is also a development team with low experience level and qualification.

Questionnaires also revealed that more than 50% of the company's projects were not able to fulfill deadlines and scope. The company's partners define their work style with absence of test planning, quality control limited to tests carried out randomly by the code programmer and customer testing at the moment when the software is delivered in the last stages of the project.

Interviews also showed that this deficient planning originates from professional experience of company partners, that is, their ideas have been influenced by values of organizational cultures of companies where they had worked previously.

Additionally, the company is relatively new in the market, has few customers and limited resources. In this way, there is a constant search for new customers and income raise, what increases the amount of projects in need for conclusion at a short time length – even supported by an inexperienced team.

To “Desktop”, company founders' previous cultural references and lack of resources to hire more experienced developers or invest in personal qualification originate unfavorable aspects to the adoption of XP, since there is low detail orientation.

5.2.2. *Favorable and unfavorable cultural aspects for detail orientation*

Due to the remarks and reflections above, it is possible to identify the cultural aspects shown in Table 4 (favorable and unfavorable to the adoption of the XP) related to the dimension Detail Orientation.

5.3. *Dimension Outcome Orientation*

The dimension “Outcome Orientation” corresponds to the level in which management focuses more on results than on techniques and results used to obtain results (Robbins, 1999).

It is important to mention Bossavit's (2002) experience in relation to this dimension. The author faced an organi-

Table 4

Favorable and unfavorable aspects for Detail Orientation

Dimension	Favorable aspects in relation to XP	Unfavorable aspects in relation to XP
Detail Orientation	– Analytical developers – Philosophy of constant improvement – Application of tests and reviews is important	– Attention to the whole – Resistance to testing and reviewing intensification – Lack of quality control

zational culture that was unfavorable to the adoption of XP method. The author denominated this culture as “do more spending less”. Bossavit comments that this culture had attitudes that made difficult his job as an XP coach.

According to Bossavit, when implementing XP project, the company did not enable the necessary conditions to supply demands required by this software development method. For example, there was no appropriate hardware available, nor adequate environment to pair program. Furthermore, great part of the more experienced developers was designated to work at other projects that the company considered more important and profitable, so that to XP project there were only inexperienced developers available.

The coach suggested a previous training for programmers who were to work at an XP project. However, the company was reluctant, and considered the training a cost to be avoided.

Bossavit faced even more resistance to adopt pair programming, which is seen as one of the “myths of pair programming” (Williams and Kessler, 2002). The myth is the idea that the practice doubles workload, since two programmers do the work of one. However, according to Williams, this practice is essential to spreading knowledge and quality of codes in XP projects. Moreover, time spent in pair programming is compensated with error reduction and eventual corrections.

This culture Bossavit identified as strongly concerned with costs is influenced by internal and external factors. Internal factors may origin from convictions and values of company members, and external factors have to do with market competitiveness or economic politics in force. Even before those adversities, it is necessary to establish a balance among profitability, minimization of costs, and other values, such as concern with employees' work conditions and software quality.

Profit-oriented companies commonly demand enhancement of developers' performance in terms of extra hours, in a way that companies may obtain maximum profit on a workday. XP has a very clarifying approach in relation

1. Are necessary material and intellectual resources made available to each software project?
2. Is there balance in relation to expected results, quality product maintenance and employees' satisfaction?
3. Do programmers use to work extra hours?
4. Do software products usually satisfy customers' requirements?

Fig. 4. Checklist items – dimension Outcome Orientation.

to this, and it is against working over a daily journey. According to Teles (2004), working for extra time is incompatible with intensive creativity required for an XP software developer.

In XP projects, the 40-Hour Week practice demands that developers work only 8 h per day (40 h per week). This demonstrates respect to each member's integrity, and project's agility increases (Teles, 2004).

Appropriate orientation to results at XP has to do with developers' motivations and availability of necessary resources to supplying customers with quality product in time. In this way, company's profitability becomes a consequence of these practices.

Thus it is necessary to verify if the software company willing to adopt XP balances profitability (and/or cost minimization) giving attention to employees, and customers. This may be observed through policies and attitudes, such as: (i) Treatment given to employees, (ii) performative demand in extra hours, (iii) incentive to developer's policies professional growth, (iv) company–customer relationship, and (v) satisfaction of the customer's requirements. Fig. 4 shows the checklist related to the dimension Outcome Orientation.

5.3.1. Scenario 3 – “Webgames” company

In relation to the dimension Outcomes Orientation, it is worthwhile saying that, at the company “WebGames”, the strong orientation to results had a negative impact on the adoption of XP method.

The company “WebGames”, which has a well-integrated and skillful team, had already adopted some XP practices in its projects, getting adapted to premises of the method. Nevertheless evolution and continuity of practices were interrupted by relevant organizational changes.

“WebGames”, which is a small company, became known with the release of a product designed by its partners. To its profile, it is an audacious and innovative project. To make business feasible, the company had to be restructured, and it was necessary to borrow capital from financial investors who work at software market.

The organizational structure of the company changed considerably, since the number of employees triplicated, and new post jobs, such as board of directors and council, were institutionalized. This result was that the company was almost exclusively dedicated to the new project, and financial resources depended on the capital of people who invested in the project.

In spite of that, investors interpreted the project merely as business, and, as such, it should be highly profitable in a short period of time in order to compensate for investments and risks. As investors commanded the company's finances, investors started to establish tight schedules and impose severe charges to results and product evolution.

In spite of the team's dedication and effort (which resulted from the group's awareness of the importance of the project to the company's survival and maintenance of their jobs), the interference of investors' confidence instituted an apprehensive environment with extrapolative workload. All in all, these factors led to the abandonment of XP practices that the group was starting to adopt.

5.3.2. Favorable and unfavorable cultural aspects for outcomes orientation

Due to the remarks and reflections above, it is possible to identify the cultural aspects shown in Table 5 (favorable and unfavorable to the adoption of the XP) related to the dimension Outcome Orientation.

5.4. Dimension People Orientation

The dimension People Orientation corresponds to the level to which managers' decisions consider their effects in relation to people at the company (Robbins, 1999).

According to Agile Manifesto (Agile, 2001), it is better to have skilful and motivated people rather than expensive technologies and tools. Highsmith (2002) pinpoints that agile methods are developers-wise, since the way software is developed depends essentially on people.

In the XP method, it is fundamental that programmers accept tasks as a responsibility, rather than a work command. This makes the method especially difficult to apply. Programmers need to feel motivated to try to think first of

Table 5
Favorable and unfavorable aspects for Outcomes Orientation

Dimension	Favorable aspects in relation to XP	Unfavorable aspects in relation to XP
Outcomes Orientation	– Valuation of employees – Concern with the customer's satisfaction – Availability of material and intellectual resources to projects	– Strong orientation to profitability and cost reduction in detriment to employees and customers' valuation – Demand developers' performance in terms of extra hours

quality and simplicity, and test continuously functionalities and the code (Beck, 2000).

To Asproni et al. (2004), software developers' motivation is influenced by intrinsic and extrinsic factors. Intrinsic factors refer to goals and individual aspirations, such as: professional realization, acknowledgement, growth expectation and social relations. Extrinsic factors depend on the scenario surrounding people and their basic needs, such as: an adequate salary, work environment and management style. It is important to highlight that both intrinsic and extrinsic factors are part of the organizational culture, and should be considered when intending to adopt the XP method.

The company interested in using XP should be concerned with promoting human capital valuing policies, and being zealous for its employees, so that employees may feel motivated enough to perform desirably according to the method. In this way, benefits and profit participation may contribute to the software development group's satisfaction.

It is also important to consider Fowler's (2003) considerations. The author affirms that, since agile processes are totally dependent on people, they cannot be imposed at the company, yet they should be widely accepted both by managers and developers.

Fig. 5 shows the checklist related to the dimension People Orientation.

5.4.1. Scenario 4 – “Web” company

The “Web” company made some information available about a project that aids on perceiving how people orientation may be a decisive factor to implement XP method satisfactorily. This could be noticed from the team and managers' perceptions and values, reported mainly during interviews in which they commented on their experiences with the project.

The company manager believes that the XP method has a competitive differential that allows for increasing productivity indicators, software quality and developers' satisfaction. Now developers demonstrated to be motivated and engaged with the XP method, and they affirmed that it represents a possibility for improving group work, increase knowledge and productivity, and raise their salaries (the company functions based on a profit share system).

The product developed in this project is a supporting tool for the competitive intelligence process. The product's target consumers are organizations looking for constant innovation and market monitoring.

Table 6

Favorable and unfavorable aspects for People Orientation

Dimension	Favorable aspects in relation to XP	Unfavorable aspects in relation to XP
People Orientation	– Motivation – Benefits – Satisfactory salary – Career plan – Good relation between the development team and the management	– Discouraged developers – Absence of human capital valuing policies – Poor relation between the development team and the management

Management fully supported the project, determining that the team should work exclusively at this activity. In this project, an employee of the company (a business analyst) represents the customer, who is also an expert on competitive intelligence.

It is important to highlight that this study did not accompany the project's development, nor refined the project's technical data (performative measures or technologies adopted). Differently, the study was restricted to identifying aspects of organizational culture.

With some research instruments (questionnaires, interviews and on-site visits), it was possible to observe that the team and the management's wide acceptance of XP and incentive policies to employees (profit share system) act favorably in respect to the adoption of XP method.

It is also important to notice that aspects related to other dimensions have also been detected, which contributed to the success of the method implementation. Some of these aspects are team experience, good relation among team members, company support to the new product, and new ways to work.

5.4.2. Favorable and unfavorable cultural aspects for People Orientation

Due to the remarks and reflections above, it is possible to identify the cultural aspects shown in Table 6 (favorable and unfavorable to the adoption of the XP) related to the dimension People Orientation.

5.5. Dimension Team Orientation

The dimension Team Orientation corresponds to the level to which activities are developed focusing more on the team than on each member individually (Robbins, 1999). In other

1. How are human resource policies like at the company?
2. How much employees are motivated and engaged with organizational objectives?
3. Are employees satisfied with their salaries?
4. Are there opportunities for developers' intellectual growth?
5. What is the relation between managers and software development team like?

Fig. 5. Checklist items – dimension People Orientation.

1. Are activities centered on individual or team tasks at the company?
2. Does the management style favor teamwork?
3. Is there cohesion among developers? Do they identify themselves as a team?
4. Do the rewarding system and company evaluation stimulate individual or teamwork?

Fig. 6. Checklist items – dimension Team Orientation.

words, it is the way the company management usually guides work tasks, either advising developers individually, or the team as a whole.

The work team is tremendously important to the method adoption in terms of evaluating, integrating and sharing collective information when executing XP practices and values. In the XP method, teamwork is extremely valued, and it becomes feasible through communication, feedback, common vocabulary and preference for more direct forms of communication.

Kirby (2003) clarifies the meaning of work group by establishing the difference between team software development and group software development. According to Kirby, a group is a team when its members interact and share work, instead of simply coexist in the same physical environment and split the project in individual parts.

When dealing with software development teams, Aspironi et al. (2004) affirms that “team spirit” requires that individuals be involved with a strong sense of identification and commitment with the team, which does not occur so easily. This occurs when all members are willing to dedicate to the team, and they are all focused on common goals leading people towards the same direction. Thus there will be a motivator to establish clear goals involving the team throughout all project phases.

Individual-oriented companies may find it hard to adopt the XP method since it discourages indirectly the “team spirit” and the shared vision of work objectives, therefore, making room for individualism.

Some company policies may also have repercussions on teamwork. In case the company’s rewarding system is turned to the individual, it may trigger competitiveness among developers, and make difficult or even prevent the “team spirit”. On the other hand, if the system in question is turned to a team, it may stimulate cohesion among developers, as well as motivation.

Another policy may echo teamwork, which is the performance evaluation system applied at the company. Individual evaluation stimulates individual performance search and may produce competition. Now team evaluation produces a sense of union, collective effort, and concern with colleagues’ difficulties, aiming at a good performance.

Fig. 6 shows the checklist related to the dimension Team Orientation.

5.5.1. Scenario 5 – “Desktop 2” company

At “Desktop 2” company, there is a good example of the difference between group work and teamwork, keeping in mind that teamwork is fundamental to the adoption of

XP method. When visiting this company, we notice that the system analyst specifies the software requirements together with the customer, and then reports them to the manager, who divides the document in modules, and determines the part each developer will receive. In turn, developers program their parts individually. As parts become ready, each one is integrated and tested.

In this scenario, the manager made a simple work division, in a way that there was no teamwork. Considering that each individual did not know much about their colleagues’ work, there was a lack of synergy and cohesion important to the team.

To use XP, developers need to be involved with a shared vision, since their activities are interconnected, and they complement each other. Developers should have a systematic vision, that is, this vision should be encompassing in a way that all implications of the task may be interconnected. In this way, it is important to notice that the company ends up impeding knowledge exchange among developers when the company lacks teamwork.

As regards to teamwork, Rising (2002) reports an experience when facing difficulties to reach goals in a project. The team improved considerably after the adoption of practices of short daily meetings. These meetings allowed developers to talk about what they were doing at that moment, and if they had any difficulties. In case they did, they would make suggestions and help each other after the meeting.

According to Rising, initially some resisted to the proposal. However, after the first meetings and observing their results, the teamwork culture was established in that group, and the project was successful.

Rising’s observations are extremely valid. Yet, when thinking of different people, feelings, realities and convictions, it is risky to think that this will occur in all projects. Thus Sommerville (2003) enumerates factors that influence teamwork. The following factors should be considered: composition (if there is balance among skills, experience and personality); cohesion (if the group sees itself as a team); communication (if the team members have communicate properly); and organization (if the team is organized in a way that all feel they are worthy and feel satisfied with their roles).

5.5.2. Favorable and unfavorable cultural aspects for Team Orientation

Due to the remarks and reflections above, it is possible to identify the cultural aspects shown in Table 7 (favorable

Table 7
Favorable and unfavorable aspects for Team Orientation

Dimension	Favorable aspects in relation to XP	Unfavorable aspects in relation to XP
Team Orientation	– Team-wise tasks – Team-oriented rewarding system – Team-oriented evaluative system – Team spirit and cohesion – Balanced skills, experiences and personalities among developers	– Individual-oriented tasks – Individual-oriented rewarding system – Individual-oriented evaluative system – Lack of cohesion in the development team – Unbalanced skills, experiences and personalities among developers

and unfavorable to the adoption of the XP) related to the dimension People Orientation.

5.6. Dimension Aggressiveness

The dimension “Aggressiveness” corresponds to the level on which people are more aggressive and competitive than cooperative at the company.

In spite of people orientation being a cultural matter, Miller (2003) states that agile methods may provide mechanisms for developers to cooperate with each other. However, the author points out the need for having a team composed by qualified professionals in order to evolve.

To Miller, to adopt an agile method such as XP, a company needs good developers who feel motivated by their team. By means of collaboration, developers may teach and guide inexperienced colleagues. Nevertheless, this cultural matter does not seem to be so simple, since it is based upon people’s convictions and values, which may be totally opposed to what is expected.

Collaboration should be involved within practices, values and the entire work environment of an XP project. Emphasis on face-to-face communication, shared work environment, interaction with the customer, and incremental and iterative development software are a few factors that serve as a proof for this premise.

In the pair programming practice, for instance, in case two people work in pair in the morning shift, in the afternoon they may easily form pairs with other colleagues. In this way, if the developer is in charge of an unfamiliar area, s/he will have to ask someone who has worked on it recently to pair with (Beck, 2000). In this case, a developer should be willing to teach the other, and share his/her knowledge.

Some signs may help to detect if there is collaboration. It is necessary to observe how relations at work take place verifying if there are team activities, knowledge share, information about tasks, and informal and clear-cut communication. Honesty, respect and trust should also exist – they are essential to the adoption of XP.

On the other hand, Cockburn (2001) pinpoints the importance of being cautious at environments where there is exces-

sive zeal and respect among colleagues to the point that developers do not disagree about technical decisions afraid of offending one. In this context, one may not take technical discussion for personal discussion. Technical discussion is positive to the project. For instance, it may take place when choosing programming language. Now personal discussion may start from personal differences that produce competition and internal conflicts during the project.

The need for a collaborative environment, in which people may trust each other, becomes very clear in the collective code practice. In this practice, any developer may modify any part of the code at any time in any part of the system.

To detect other programmers’ errors and alter their codes, the team members should be honest and respectful with each other, and also courageous to correct what is wrong, even a colleague’s error. Developers need freedom to express their fears, and they need to be supported and support others.

Collaboration should stimulate by means of rites of integration, such as fraternizations and sports practiced in groups. Even though there are divergences and conflict among employees, these practices reinforce association, and soothe any possibility lack of collaboration.

Differently management may also stimulate competition among developers signaling the possibility of obtaining rewards and individual acknowledgement. However team members themselves may create a competitive atmosphere, in case that is the developers’ profile. In this case, competition may arise as ego dispute. Even so, competition among group members is unfavorable to the adoption of XP.

Personal conflicts in the development team are also unfavorable to XP. It would be very difficult to put in practice pair programming and collective code among developers who have emotional constraints in relation to the other.

Fig. 7 shows the checklist related to the dimension Aggressiveness.

5.6.1. Scenario 6 – “Departamento” company

The analysis of the information technology department of “Departamento” company allowed for the identification of some important aspects of its organizational culture related to the dimension aggressiveness.

At this department, some developers affirmed to have some interest on the adoption of XP, yet they knew that the manager’s values would be one of the main hindrances. To them, the manager’s personal values are based on professional ascension and power in order to reach higher posts at the company. These values lead attitudes which the team reproves, such as superestimating his/her own merits and achievements in the project in detriment of colleagues’ qualities and deeds. This behavior seems to be based on the manager’s belief of strengthening within power hierarchy established at the company.

These aspects produced dissatisfaction and suspicion at the department to the point that colleagues started to see each other as rivals who may threaten other team members’

1. Do developers usually run to colleagues when they face doubts or difficulties? Is there mutual help?
2. Do developers share knowledge and information?
3. Is there integration among development team members?
4. Is there competition among developers?
5. Are there conflicts among developers? How are they solved?
6. How is the relation between the team and the management?

Fig. 7. Checklist items – dimension Aggressiveness.

Table 8
Favorable and unfavorable aspects for Aggressiveness

Dimension	Favorable aspects in relation to XP	Unfavorable aspects in relation to XP
Aggressiveness	– Collaboration – Knowledge and information sharing – Trust and informality – Solving conflicts democratically	– Competition – Formality – Personal conflicts – Solving conflicts through imposition

position. In fact, from talking to other team members, we notice that this power dispute and competition was not only limited to the manager, but it was spread over the company's organizational culture.

In this case, it seems improbable that XP practices may offer efficient mechanisms to generate collaboration and modify the organizational culture composed by rooted values and convictions which determine certain behaviors observed at that work environment.

5.6.2. Favorable and unfavorable cultural aspects for Aggressiveness

Due to the remarks and reflections above, it is possible to identify the cultural aspects shown in Table 8 (favorable and unfavorable to the adoption of the XP) related to the dimension Aggressiveness.

5.7. Dimension Stability

The dimension “Stability” corresponds to the level on which activities focus on maintaining status quo (Robbins, 1999).

A company's culture may not be adaptable but focused on status quo to the point that it shows conservatism and a low capability for change. This type of company keeps ideas, values and habits which remain rooted and do not change with time, that is, they remain as if nothing had changed happened beyond the company's boundaries (Chia-avenato, 1999).

Adopting XP method is a considerable change for any company. However some organizational elements relative to the dimension of Stability may be a barrier to adopt XP method. Change always brings problems independently of the level of planning. However, the quantity and types of problems depend on the company and its capability of changing (Astels et al., 2002).

Some companies are not very dedicated to adapting to the market and new technologies, attributing more value to predictability and security already obtained through techniques that have been successful at the company with time.

Other companies are concerned with incorporating new technologies and comprehending market tendencies. Generally companies admit the need for adaptation, but they do not make an effort to make it real. In this sense, it is necessary to verify if the company is used to the adoption of new technologies, new work manners and especially identify who should be mobilized within the organization structure and the type of influence they may have on the adoption of XP method.

Difficulties on accomplishing changes may be differently interpreted by varied points of view of people who are part of the organizational structure. For instance, a company investor's or director's motivations may be different from a programmer's.

Fig. 8 shows the checklist related to the dimension Stability.

5.7.1. Scenario 6 – “WebGames” company

“WebGames” company, which has been mentioned on item 5.3.2 on Outcomes Orientation, also presents a scenario related to the dimension “Stability”.

At “WebGames”, the adoption of XP method could be interpreted as a competitive differential to increase business profitability. However, investors and directors who misinterpreted it as a change that would bring instability and risks blocked initiatives to adopt the method.

This attitude reveals that resistance to change is a way of staying at a comfort zone, avoiding new situations that would alter work routine, and consequently provoke risks.

1. Is the company open to trying new technologies and different ways to work?
2. Does the company usually design new products?

Fig. 8. Checklist items – dimension Stability.

Table 9
Favorable and unfavorable aspects in relation to the adoption of XP

Dimension	Favorable aspects in relation to XP	Unfavorable aspects in relation to XP
Stability	– Open to new technologies – Open to new ways to work	– It is limited to known technologies even without satisfactory results – It seeks for security in routine techniques even without satisfactory results

On site observations also shown that some of the analyzed companies seek for customers as a priority, and increasing their market participation. Thus, investments on improvements, such as software development process reevaluation, come on second place. Besides cultural matters, it was observed that this kind of behavior tends to be more present in small and relatively new companies that are trying to find their share in the market, and becoming profitable.

5.7.2. Favorable and unfavorable cultural aspects for Stability

Due to the remarks and reflections above, it is possible to identify the cultural aspects shown in Table 9 (favorable and unfavorable to the adoption of the XP) related to the dimension Stability.

6. Final remarks

Many authors have presented XP as a feasible and advantageous alternative to software companies with small teams working with vague requirements. If adequately applied XP can provide good results, such as agile projects and code quality.

Nonetheless, literature on XP generally assumes the existence of an environment at which motivated and collaborative developers work as a team, developing, fully testing and reviewing code.

In this context, this study aimed at highlighting the various organizational aspects necessary to practice XP satisfactorily. However, these aspects are not always present at software companies. Furthermore this study pinpointed that it is possible to trace aspects that hinder the method implementation.

Empirical observations mentioned throughout this study corroborate the existence of cultural aspects at the software companies analyzed, as well as they demonstrate that these aspects may facilitate or hinder the adoption of XP. In this way, the review of the literature also confirmed an existent relation between the company's culture and a satisfactory adoption of XP.

Software companies may use the results of this study, above all the identification of favorable and unfavorable aspects to the adoption of XP, as a parameter for preliminary remarks on the convenience of XP implementation. In other words, companies may check which favorable and

unfavorable aspects pointed out in this study are present in their organizational context, and then evaluate previously the feasibility of XP adoption.

The Checklist (applied in form of questionnaires, interviews, and local observations) is also a mechanism that can be used by software companies since it aids to reveal each company's cultural aspects.

These remarks enable companies to evaluate the depth of necessary transformations to the adoption of XP, as well as to analyze if it is possible to remove culture barriers eventually detected.

Another important remark refers to the importance of interviews and the insertion of an observer at the company analyzed. Questionnaires alone as a means to comprehend the company's culture seem not to be sufficient to grasp subjective information as those.

Thus Schein's (1999) considerations that each company pursues its own profile with cultural assuredness, which would escape from any questionnaire, are very pertaining. To Schein, there may be problems when interpreting answers. Some employees may answer reflecting their fear for being reproached by managers in case they answer negatively, and they may answer according to a desirable culture, which does not actually represent the company's culture.

To conclude, investigations and remarks of this study may explain reasons why companies are unable to adopt XP throughout, and thus remain restricted to some XP practices, even aware of benefits brought by XP method. While some practices encounter a favorable environment at the company with the necessary cultural aspects, some others encounter hostile cultural aspects for their implementation.

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